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Editors

Neural Mechanisms

New Challenges in the Philosophy
of Neuroscience

 Springer

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Chapter 19

What Is the Job of the Job Description Challenge? A Study in Esoteric and Exoteric Semantics



Colin Klein and Peter Clutton

Abstract Ramsey’s *Job Description Challenge* has received substantial attention in debates over mental representation. We distinguish two senses of representational semantics: *esoteric* semantics, which concern the relationship between a representation and the world, and *exoteric* semantics, which concern the semantics of representations within a computational framework. Given that pair, we argue that there are three ways in which you could look to cognitive science to answer the job description challenge. Two of those ways make the job description challenge trivial—that is, answerable but uninteresting. We think that the recent literature has focused on readings to the challenge that possess this failing. We argue that Ramsey’s challenge is best understood in terms of the interaction between esoteric and exoteric semantics. This third reading is more complicated to address but more interesting to answer. Understood in this way, the answers to the challenge will be local and case-by-case. On some of these cases, as we review, the challenge is met.

19.1 Introduction

In an influential passage, William Ramsey sets out what he calls the *Job Description Challenge* for representationalist theories in cognitive science. If we want to understand certain cognitive processes as representational, then

... we need to be told, in presumably computational, mechanical or causal/physical terms, just how the system employs representational structures. Principally, there needs to be some sort of account of just how the structure’s possession of intentional content is (in some way) relevant to what it does in the cognitive system. After all, to be a representation, a state or structure must not only have content, but it must also be the case that this content is in some way pertinent to how it is used. We need, in other words, an account of how it actually serves as a representation in a physical system; of how it functions as a representation (2007, p. 27)

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Positing representations, in other words, must do some useful work for cognitive scientists; conversely, if nothing empirical turns on whether something is a representation or not, then naturalistically inclined philosophy of mind ought to avoid the term.

One way to read the job description challenge is as asking whether explanations in terms of representation involve something over and above mere “intentional glosses” (Egan 2014, p. 128). The question is then whether that intentional explanation is “part of the essential characterization of the device” or simply “ascribed to facilitate the explanation of the relevant cognitive capacity”, that is, simply to make clear “how the computational/mathematical theory addresses the intentionally characterised phenomena with which we began and which it is the job of the theory to explain” (Egan 2014, p. 128).

This debate still animates philosophy of cognitive science. We will focus on debates where specific sorts of representations are invoked. To take a recent example, consider Gadsby and Williams (2018)’s argument that the cognitive neuropsychiatric theories of body representation and misrepresentation provide a good case against the standard anti-representationalism of Hutto and Myin (2012). They argue that “a robustly representational concept—the body schema—is explanatorily central within this research” and that these representations have “satisfaction conditions of some kind”, allowing them to “misrepresent” (p. 5298). Neander (2017), in another recent example makes a similar case regarding cognitive neuropsychological explanations of certain visual deficits. Again what is at issue is that “what is posited is intentional, insofar as the relevant mental content permits the possibility of error and hence is not mere (i.e., natural-factive as opposed to intentional) informational content” (Neander 2017, p. 27).

Both Neander and Gadsby & Williams explicitly address Ramsey (2007) in this line of argumentation. Ramsey argues that naturalism entails that in some sense, representational explanations are not completely necessary, and that the use of representational explanations in cognitive science would be cause for representational realism only when those explanations provide non-trivial explanatory purchase. Neander and Gadsby & Williams in their arguments aim to show that indeed this is true of their respective case examples—the representational explanations do provide non-trivial explanatory purchase—and draw representational realism conclusions from that. If these fields are on the right track, then we have good reason for representational realism.

Note that in each case, the Job Description Challenge is typically accepted as legitimate, and responses try to show how the challenge is met by the practice of ordinary cognitive scientists.

We think that there is a problem with the Job Description Challenge itself, at least on some readings of it (and these readings have received a lot attention). In what follows, we will distinguish three different readings of Ramsey’s challenge. Two of those readings, we will argue, make the Job Description Challenge relatively uninteresting. The final way makes the challenge more interesting, but also requires local and case-by-case responses. On some of these cases, we argue, the challenge will be met. More broadly, we suggest, the delineation of different readings of the

Job Description Challenge shows how distinct readings have been confused, and generated philosophical puzzles beyond what is necessary.

19.2 Two Types of Semantics for Representations

The attraction of representationalism in cognitive science arguably derives from the attraction of computationalism. Conversely, many of the same issues about representation occur in the computational literature (Piccinini 2008). So to begin, let's start with a relatively concrete computational example. Suppose I'm writing an app to help tourists navigate Canberra's sprawling light rail system. I will need to represent the stops and their relationship to one another. *What* I need to represent is thus fixed by the world plus what I need to do. I still have considerable latitude in *how* I represent that information. I might use a list of stops, or a directed graph, or a set of tuples, or a matrix, or any of an indefinite variety of other data structures.

The choice of data structure should be driven, in part, by what I want to *do* with the information I represent. As Marr puts it:¹

...even though one is not restricted to using just one representation system for a given type of information, the choice of which to use is important and cannot be taken lightly. It determines what information is made explicit, and hence what is pushed into the background, and it has a far-reaching effect on the ease and difficulty with which operations may be subsequently carried out on that information. (1982, pp. 21–22)

Different data structures have different properties. These determine which algorithms can be run on them, and how efficiently they can be run. Philosophers of psychology often talk about tasks being performed by algorithms *tout court*, but strictly speaking an algorithm can only be specified relative to a (possibly abstract) data type (Wirth 1976). The algorithm for finding the next stop given an index might be simple if we represent stops with a list, but require an exhaustive search if we use a set of tuples.

This is not merely a point about programming. As Pylyshyn (1984) perceptively notes, cognitive science cares not just about weak, input-output equivalence but also a strong equivalence in terms of algorithms. In practice, this can often be determined by things like relative timing data or resource usage.² The algorithms available to solve a computational process, and the efficiency with which they run, are dependent on the datatype used.

¹Ritchie (2019) notes that Marr's classic presentations of computational analysis repeatedly recognized a duality between algorithmic process on the one hand and representation on the other, and that this duality is present at each of the classic levels of analysis.

²Or in the most general sense, by the computational complexity of different algorithms. Aaronson (2015) provides a nice introduction of complexity pitched at philosophers. Complexity profiles show how various parameters scale with input size, which is why relative scaling is often useful evidence in cognitive science.

Marr's point about what is 'made explicit' and what is 'pushed into the background' is also important. A list emphasizes the ordering of stops, as well as embodying a certain pessimism about the likelihood that the system will expand beyond a single line. A graph emphasizes connectivity, and builds in optimism and future flexibility at the price of added complexity. A large map-like matrix might emphasize spatial layout: this makes connectivity difficult to extract but it's easier to link up to other maps. And so on. Each may carry the same information about the target domain, but are more or less difficult to use for different purposes. All of these are utterly familiar sorts of tradeoffs to programmers.

Suppose we settle on using a list. Following Cantwell Smith (1996, 33ff), we note that there will be two kinds of questions one can ask about that list. *Exoteric* questions about representations concern whether and how the representation we chose hooks up to the world. Does it make sense to treat our list as a list of *stops*? Or is it really a list of *shops*? Does it have the right sort of systematic relationship, or use by internal consumers, or selection history, or whatever to really *represent* the light rail line? Note that we might think that a mere list falls short of whatever my brain does, and so my app does not reach full-fledged representation. But it still makes sense to ask exoteric questions.

Esoteric questions, by contrast, are those internal to the logic of programs and the computations they give rise to. Esoteric semantics determine what it means to say that I am using a *list*, rather than some other data structure: that is, what it means to *be* a list. Esoteric semantics are thus tied up with what expectations I can have about lists when I manipulate them. Is a successor always defined? Is my representation the sort of thing that you can sort, concatenate, and duplicate? If I access it twice in a row in the same way, am I guaranteed to get the same result? The answers to esoteric questions are determined by the semantics of my programming language, not about the world.

Note, following Smith's (1996) critique of Fodor, that esoteric questions are still questions about *semantics* of programming languages (and so ultimately about the computational objects that they designate) rather than the syntax of expressions in a programming language. Syntax affects semantics, of course, but important aspects of esoteric semantics outstrip syntax. So, for example, the syntax of Python tells you that `string_one+string_two` is a well-formed expression, but nothing at all about the operation of string concatenation that is indicated by that expression. We are concerned with the latter: that is, with data structures and operations on them, rather than the syntax of expressions which refer to data structures and operations.

Esoteric semantics thus detail a set of *guarantees* about what operations I can perform on instances of that datatype. As a standard textbook puts it, datatypes are simply "... defined by some collection of selectors and constructors, together with specific conditions that those procedures must fulfill in order to be a valid representation" (Abelson et al. 1996, p. 91). These 'specific conditions' are relative to operations on the datatype itself, not the relationship between data and the world.

Guarantees might include not just what can be done to a datatype, but how efficiently those operations can be performed and what sort of resources different operations might require. Datatypes are defined and distinguished by their guaran-

tees, which is why details of algorithms depend on the datatypes upon which they operate.³

Esoteric and exoteric conditions clearly come apart. Consider maps. Rescorla (2009) makes a useful distinction between representing geometric structure and *replicating* geometric structure. Concrete maps represent geometric structure by replicating it, but there are numerous possible ways to represent the same geometric structure. Hence there are esoterically different datatypes which can play the same exoteric role. Conversely, there are things which meet the esoteric conditions for maps without representing anything at all. A map of Middle-Earth is still a map, precisely because it has a spatial structure of the right sort. Hence “replicating” geometric structure cannot mean simply mirroring a geometric structure which actually exists: to have a geometric structure is instead to meet certain esoteric guarantees (density, the triangle inequality, etc.) that maps must have to be maps.

Both esoteric and exoteric semantics come equipped with their own variety of normativity. On the exoteric side, my list *misrepresents* if it fails to line up with the world in the right way: if it misses stops, or gets the order wrong. A list which misrepresents might still be perfectly good as a *list*, though. Conversely, on the esoteric side, there is a straightforward sense in which my list can misfire as a representation by failing to live up to its guarantees. My list might become corrupted, or fail to return the right things in the right order. Any of these would count as a form of ‘miscomputation.’ Note, however, that not all exoteric failures need be esoteric failures: my list might be deficient *qua* list while still being usable for the purposes required. Conversely (and this surely happens), my list be slightly unreliable in fulfilling its guarantees while still being reliable enough to satisfy the exoteric conditions on representation.

19.3 Structure and What This Is Not

We speak of a representations, considered as instances of a datatype, as having a *structure*. But it is important not to confuse the esoteric structure of a list with either the structure of an implementation or the structure of the thing it is used to represent.

On the one hand, the thing which *implements* a list need not have the structure of a list (except in the completely derivative sense that it is an implementation of a list). My list of station names has a linear ordering. But what implements it need not, and often will not, have anything like a linear ordering. Different entries on a

³“We must decide in each case how much structure to represent in our tables, and how accessible to make each piece of information. To make such decisions, we need to know what operations are to be performed on the data. For each problem considered in this chapter, therefore, we consider not only the data structure but also the class of operations to be done on the data; the design of computer representations depends on the desired function of the data as well as on its intrinsic properties. Indeed, an emphasis on function as well as form is basic to design problems in general” (Knuth 2011, Volume 1, p. 238).

list can reside in arbitrarily dispersed parts of memory. If it's a very long list, some might be in memory and some swapped out to disk. What makes this a list, in the esoteric sense, is precisely the operations which I can perform on it.

A list is thus an odd sort of object, metaphysically speaking: what it takes to count as a list is entirely defined by the esoteric semantics, and those merely pick out a collection of things that you can *do* to a list. This is true of datatypes in general. Indeed, although there is a sense in which persistent data must be stored as something which persists, the structure of data itself need not be implemented by anything conceptually static.

Consider, for example, Abelson, Sussman, and Sussman's demonstration of different ways to implement an ordered pair. They note that one could store each item in an ordinary variable. One could also implement a pair purely functionally, by using a pair of functions which return specific values when invoked along with a setting function which creates new functions as needed. This procedural representation, they note, "... is a perfectly adequate way to represent pairs, since it fulfills the only conditions that pairs need to fulfill" (1996, p. 92). Thus there is an important sense in which a 'representation' like a list need not involve anything 'object'-like at all, because the relevant guarantees might be met purely functionally. That functionality must ultimately be cashed out in physical stuff, but the structure of the stuff needn't bear any straightforward relationship to the esoterically defined structure.

There is also an important distinction between our sense of esoteric structure and what has come to be known as *Structural* or *S-representations*. Ramsey-style S-representational accounts understand "cognitive representations as internal, structure-preserving models or map-like mechanisms." (Lee [Forthcoming](#); see also O'Brien and Opie 2004; Ramsey 2007 3.2ff). A good S-representation preserves structure which can be exploited to solve particular tasks (Gładziejewski and Miłkowski 2017). The success or failure of a representation depends on the degree to which a representation resembles its target in exploitable ways (Lee [Forthcoming](#)). As Ramsey puts it, structural representations can function as models, and are useful when there is "a type of isomorphism between the sketch and the target that can be exploited to learn certain facts about the target" (Ramsey 2007, 82). There has been considerable debate about whether all neural representations are of the structural sort, and whether proposed conditions are too strict (Shagrir 2012; Morgan 2014). We think that S-representationalism has important insights.

Yet we again emphasize that the esoteric sense of 'structure' is fundamentally a matter of the constraints placed on the datatype, not the particular instances of data represented. The structure of a list is determined by the esoteric guarantees on lists. One can concatenate lists, or create functions which return a new list by copying the original and removing the first element. You can't do that with train lines. Just as the format of a representation can diverge arbitrarily from the format of the thing

represented, so too can the (esoteric) structural demands on a representation diverge from the (exoterically relevant) structures that are represented by that item.⁴

Of course, it is not a mystery that the structure of a list is particularly handy for representing a linearly ordered set of stops: the easy operations on a list are ones that make exploiting it as an S-representation particularly handy. As a different way of making the point, we note that the structure of a datatype is relevant to other, counterfactual uses of the same data. Computational libraries, for example, often make available particular kinds of structured data. That structure is relevant insofar as it permits and constrains other, as-yet-unconceived, uses of the same bits of data. Hence the relevant notion of esoteric structure is one that *must* be able to vary independently of the exoteric uses to which data is put.

19.4 Two Uninspiring Versions of the Job Description Challenge

Having made the distinction between esoteric and exoteric aspects of representations, we return to the job description challenge. Recall that the job description challenge required a story about how representations are useful to cognitive science. We use the esoteric/exoteric distinction to suggest that this question can be disambiguated three ways. Two of those ways, we will argue, provide a response to the job description challenge, but a fairly trivial and unsatisfactory response. We clear them away to focus on a more interesting case in the following section.

First, one could read the Job Description Challenge as focusing on esoteric aspects of representations *alone*, in the absence of any further exoteric questions. That is, one could read it as asking whether thinking about vector operations or syntactic tree structures or maps could be explanatorily useful. Since the irrelevance of exoteric links to the world is presupposed here, the usefulness would have to come down to the usefulness of particular data structures in supporting various kinds of computations.

This is an unusual way to think about the Job Description Challenge, in part because it makes the challenge too easy. Arguments about the computational usefulness and necessity of particular kinds of data structures abound in cognitive science. This is, arguably, simply because these things are so useful in computer science, and that usefulness carries over to computational explorations of the mind. Inverting the question, a putative data structure earns its keep precisely *because* of the explanatory work it promises to do. On this reading, then, the Job Description Challenge is satisfied, though that success is unlikely to surprise anyone.

⁴For a neural example, see Goddard et al. (2018), who examine this issue in the case of dimensionality reduction of single-cell recordings. They note an important distinction between the structure of feature spaces and the structure of representational spaces, and suggest that apparently conflicting results about population coding can be reconciled by careful distinction of the two.

Second, one can read the challenge as focusing on exoteric conditions *alone*. So suppose that some exoteric conditions are wholly distinct from esoteric conditions. That is, it takes something for a cognitive item to count as a real, full-fledged representation, and that something doesn't have anything to do with the esoteric conditions that make it the computational sort of representation that it is. Whatever these additional exoteric conditions are, we can pose the job description challenge with respect to them: what does this additional stuff do, and do cognitive scientists care?

We suspect that this is how a lot of philosophers think of the Job Description Challenge. It is what generates staple puzzles about Twin Earth and Swamp Man. Those are cases where all the esoteric conditions are met, if they are discussed at all; it's only some aspect of exoteric conditions that are missing.

Many philosophers seem to read the challenge in this way. They look to cognitive science practice to find out whether scientists use the notion of exoteric error to answer the challenge. Where they find evidence for it, they claim that the challenge has been met and that representational realism follows. Thus there is a kind of *methodological* argument meant to settle a question about the existence of representations. Indeed, in one of the examples mentioned earlier, Neander explicitly frames her overall argument as a methodological argument: from the practice of cognitive scientists, we can draw conclusions about representations. That is, when the practice of cognitive science includes the use of representational explanations that provide 'non-trivial explanatory purchase' (2017, p. 85), we have real cause for accepting representational realism.

Yet we have also come to find this way of putting the challenge a bit puzzling. The interesting questions cannot simply be about whether cognitive science provides explanations that use explicitly exoteric terms. Everyone agrees that cognitive scientists themselves talk about representation in this sense all the time. Nearly all the answers to 'how does an organism do thus-and-such' will be cashed out in terms of error-supporting representations. Given such a question, there is an explanation in exoteric intentional terms that involves the possibility of error relative to the task at hand. This is because the task description itself is typically given in intentional terms: something like 'how does the organisms distinguish (these) EDGES from (those) SURFACES?', and that only makes sense if failure is possible.

This is especially true of cognitive neuropsychology, which was drawn on by both the Neander and Gadsby & Williams papers we cited at the outset. Cognitive neuropsychology is explicitly built around the assumption that a particular task of interest can succeed or fail to be performed (see, for example, Coltheart 2001). A typical starting point is to ask *how do we represent this thing and how can we go wrong* (see, for example, Striem-Amit et al. 2018).

If this is your picture of how the science works, then it shouldn't be a condition on the naturalist picture that it sort out the exoteric semantics. Because, in an important sense, it's a background *presupposition* of doing cognitive science in the first place that there's some useful way to do so. We take this attitude to be exemplified by Chomsky; consider, for example, his remarks that:

Thus I understand “mental” to be on a par with “chemical”, “optical”, or “electrical”. Certain phenomena, events, processes and states are informally called “chemical” etc., but no metaphysical divide is suggested thereby. The terms are used to select certain aspects of the world as a focus of inquiry. We do not seek to determine the true *criterion of the chemical*, or the *mark of the electrical*, or the *boundaries of the optical*. I will use “mental” the same way, with something like ordinary coverage, but no deeper implications. By “mind” I just mean the mental aspects of the world, with no more interest in sharpening the boundaries or finding a criterion than in other cases. (1995, p. 1)

In other words, cognitive science *begins* by assuming that there is a useful notion of representation to be had, and investigating the conditions on it. Giving exoteric semantics is useful, but it is fleshing out a presupposition that is already there.

Indeed, Ritchie (2019) suggests that a similar view is present even in Marr. On Ritchie’s account, representational content is actually part and parcel of Marr’s computational level. As he notes, such an inclusion can “be motivated on more principled grounds by considering that representation and process are core to the very idea of an information-processing task, and Marr’s levels are supposed to explain different aspects of how a system carries out such tasks” (2019, p. 1087). Hence even in classical presentations, some sort of exoteric validity is a foundational assumption.

Indeed, we think that a purely exoteric reading might end up making the Job Description Challenge seem needlessly difficult for the naturalist. The challenge appears to involve methodological deference to the natural sciences on the one hand, combined with a claim that scientists themselves might be systematically wrong about the core explanatory concepts they presuppose. It thus invites the naturalistic philosopher to take up a stance external to cognitive science and decide whether certain practices live up to an additional, extra-scientific set of criteria. Understood that way, the naturalist ought to simply reject the challenge.

19.5 Combining Esoteric and Exoteric Conditions Together

The first two ways of reading the challenge made a relatively sharp distinction between *whether* and *how* something represents the world—that is, between exoteric and esoteric. We argued that if the job description challenge is addressed to one of these questions in isolation, it fails to be compelling.

A sharp line between the two questions, however, is a philosophical artefact. We suggest that in cognitive science, the questions are usually treated as interacting. That is, it is a general assumption that one can’t figure out what a particular bit is representing without *also* knowing some things about the nature and structure of the representation itself.

On this reading, the job description challenge may be read as demanding to know whether the particular combination of esoteric and exoteric criteria employed by cognitive scientists are actually useful enough to continue employing them. Note

that this is no longer a general question: it depends on particular uses of particular representational posits.

We think that this version of the Job Description Challenge is both non-trivial and a naturalistically respectable question. We first review some examples of the approach in cognitive science, and then step back to think about how these explanations might answer a more robust and interesting version of the job description challenge.

For example, consider recent debates over the representation of the space near the body. The brain treats the space near the body in special ways. Graziano and colleagues (1994) showed that the precentral gyrus of macaques contained neurons with bimodal visual and tactile receptive fields which densely overlap near the body (Graziano and Cooke 2006, fig. 4). Stimulation of the same neurons can evoke complex defensive behaviors directed towards near space (Graziano et al. 2002a, b; Graziano 2006). Reviewing these findings, Graziano and Cooke (2006, p. 846) suggested that “a major function of these cortical areas is to maintain a margin of safety around the body and to coordinate actions that defend the body surface.” This margin has come to be known as *peripersonal space* (PPS).

There is an active and vigorous debate about the particular properties of PPS representation. A parallel debate concerns whether we really represent *peripersonal* space, or if we merely represent (e.g.) potential actions in space, and the space near our body happens to be a place where many of our interests coincide. Grush, for example considers neurons in these areas to underly the “capacity to represent an environment of actionable objects” (2007, pp. 342–43), rather than an action-neutral space.

What is useful about this debate is that it shows how exoteric and esoteric questions are deeply intertwined. One cannot separate the questions of what PPS representations refer to from the question of how many there are, and that in turn depends on questions of what format that they might be in.

De Vignemont and Iannetti (2015), for example, argue that there must be two (and only two) distinct representations of PPS. They argue this on the basis that certain tasks distort PPS representations, and that the distortions for defense and for tool use affect PPS in different ways. Klein (forthcoming) responds with a model which accounts for the data using a single representation with a more complicated coordinate basis. The debate between the two, as Klein notes, depends on whether the format of PPS representations is a low-dimensional cartographic representation or a higher-dimensional space with a nonlinear basis.

In a recent review, Bufacchi and Iannetti (2018) similarly suggest that PPS is represented by a series of ‘fields’. These fields, like physical fields, take a scalar quantity at every point in space. Noel and Serino (2019) respond that a representation that has continuous coverage of an indefinite region of space is biologically unrealistic. In response, Bufacchi and Iannetti (2019) argue that the representation is functional: what is important is that PPS representations allow the brain to systematically recover and transform action values given a specific point in space. Thus while a dense matrix would be unrealistic, there are other lightweight ways to meet the necessary guarantees.

This strikes us as paradigmatic cognitive science. In each of these debates, note, there is a crucial interplay between esoteric and exoteric semantics: whether something represents, and what is representing, is taken to depend crucially on *how* it represents. Esoteric semantics demands stories about plausible formats and plausible operations that can take place on those formats, while the exoteric part constrains those stories by the external role that the representation plays. Klein, for example, points out that if one is restricted to linear readout functions, then only a nonlinear basis will allow for a unified representation of PPS in the domains that De Vignemont and Iannetti care about.

This reading of the job description challenge takes seriously Ramsey's demand to know "just how the system employs representational structure". If esoteric and exoteric semantics partially interact, then whether and how a representation is fit for purpose will depend both on the details of that representation's structure and how that structure is used by the cognitive system.

Thus the job description challenge is not, first and foremost, something that can be solved once and for all for cognitive science, at least in any substantial way. (This is another parallel with computer science, where there is relatively little discussion of representation as such, as compared to endless discussion of different representational structures and their merits.) Instead, the job description challenge is fought case-by-case.

Of course, on this reading, there remains the possibility of interesting debates about exoteric semantics. Indeed, it is at least possible that we might end up denying that some esoterically respectable representation really counts as a representation.

So for example, Barbara Webb (2006; Webb and Wystrach 2016) suggests that many of the computations in insect navigation should be counted as mere transformations, rather than representations of space. Her argument turns on fine-grained details of the properties of the underlying computations, and is done against a background against which some neural computations *are* supported by representations. Barron and Klein (2016), by contrast, suggest that the integrative action of the central complex is sufficiently complex to elevate these same procedures to representational status.

Whichever side you fall on in the insect navigation debate, we suggest, this debate shows that the job description challenge is met. "Insects don't represent space because they use a specific kind of landmark-based pattern matching combined with gradient descent" is a substantive and scientifically fruitful hypothesis; pushing back against it has required detailed discussion of the neural details.

Indeed, we note that among the anti-representationalists, there is considerable variation on this very question. When Van Gelder (1995) introduces a dynamic, anti-representationalist approach to cognition, he is fairly clear that he is presenting an empirical *alternative* to traditional computational representations. In our terms, he posits entities with radically different esoteric conditions, and suggests that these can be conjoined with exoteric conditions which meet this challenge. Chemero (2009) similarly takes the interesting fight to be wherever he and traditional computationalists agree on exoteric questions, and argues instead about whether

alternative dynamic notions might be explanatorily fruitful.⁵ We are (by and large) representationalists, but we think this form of anti-representationalism is playing on the naturalistic grounds that the Job Description Challenge demands.

We have said that there is a background presupposition regarding the use of representations in many areas of cognitive science. For that reason, we discouraged the use of easy arguments from scientific realism to realism about representations. Explanations that advert to representations come cheaply in many areas of cognitive science. But that does not mean we endorse the kind of instrumentalist reading of the scientific practice of representation that has sometimes been offered (Chemero 2009; see also Lee 2018). We encourage instead a way of looking at these practices that asks questions with substantive empirical force, as in our example regarding the nature of the computations used in insect navigation.

Our position also does not choose sides in the standard breakdown between ontological and methodological naturalism (Caiani 2018). Ontological naturalists about representation looks for certain types of physical objects and properties to play the role that representation plays; methodological naturalists look to the explanatory utility of representation in our best scientific practices.

Our position doesn't stake out a claim on this familiar territory in any straightforward way. We have said that on certain ways of looking at this question, the use of representation in explanation, the methodological side, is more of a presupposition than the kind of practice that ought to be used to verify the presence of representation in any way. And further, that on the ontological side, there is again a type of trivial answer in the area that of course there are naturalistic structures that play an interesting role in cognition when performing various tasks. Where there are interesting questions, we have said, they will require examining particular combinations of esoteric and exoteric criteria employed by cognitive scientists and deciding whether those particular combinations are useful enough to continue using.

This is the most interesting reading of the job description challenge. It poses a fruitful, substantive question, the answer to which has both scientific and philosophical import. And, at least sometimes, this more substantive version of the job description challenge can be met. That is an interesting result.

19.6 Conclusion

Naturalistic challenges in philosophy always walk a fine line: they must balance what philosophers think scientists should care about with what scientists actually

⁵“I take it that using the newer, more restrictive definition to try to argue in favor of nonrepresentational cognitive science would be problematic. ‘Using my new definition of representations, none of these systems has representations’ is a near neighbor of the Hegelian arguments deplored [earlier]. That is, it allows radical embodied cognitive scientists or their opponents to win arguments by re-defining terms. For purposes here, then, the traditional views are more appropriate . . .” (Chemero 2009, 66).

care about. In the case of representation, much of the philosophical interest comes from the putative power of representational theories to solve old problems about intentionality and the nature of the mental. Yet solving those problems is not the reason why representations appear in empirical explanations, and the surrounding apparatus of cognitive science was built to tackle very different questions.

The Job Description Challenge is posed in a naturalistic spirit. We suggested that the reading which combines esoteric and exoteric conditions on representations is faithful to that spirit, and is interesting enough that it is the grounds for meaty fights between representationalists and anti-representationalists.

Why all the heat and noise, then? We conclude with a tentative diagnosis. We noted that focusing on either esoteric or exoteric questions in isolation was relatively uninteresting: indeed, they make the Job Description Challenge itself seem like a mistake. There is an understandable philosophical tendency to break difficult problems down into their component parts. Furthermore, keeping one part of a problem fixed while investigating another is, for many philosophical problems, the best way to get general solutions.

So, for example, much of the description of ‘pure’ exoteric problems ends up bracketing questions of esoteric structure: internal representations might as well be blinking lights. But then all that one can say is that of course cognitive science cares about representation: look how often they talk about it! Bracketing exoteric questions leads to a similarly unproductive sort of stalemate. So the Job Description Challenge seems like it ought to have bite—yet many ways of actually trying to approach it end up solving a much less interesting problem.

That is perhaps what should be expected. The interesting questions about representation, if the above is correct, are primarily *local* questions: we can ask about whether this or that way of dealing with peripersonal space is a representation of PPS, and in what sense it is. This depends in intimate ways on both the structure of the representation and the domain of the representation, however, and there is comparatively little that carries over to a discussion about how insects navigate the world.

Splitting esoteric and exoteric questions thus creates confusions without delivering generality. We take it that the main contribution of our paper is distinguishing two types of question that have often been considered separately. We have done so, however, in order to warn against pursuing them separately.

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